

been all about primary automation and sensors, IoT devices of the future will be integrated deeper and possess more capabilities on the edge. The impact of 5G can be divided into three major areas within IoT: home automation and security, automotive and healthcare. Here's how they'll benefit specifically:

- **Home automation and security:** While the capability to handle more devices is already a plus for home automation, 5G's low-power, wide-area network [LPWAN] capability is more lucrative for IoT applications, increasing the battery life of IoT devices at home, as well as reducing connectivity hassles. With LPWAN, a 5G device will be able to connect to the internet directly without depending on home WiFi to do so.
- **Automotive:** Summing up the impact of 5G on the automotive industry in a few lines won't do it justice. One look at CES 2019 would be enough to see demonstrations from numerous companies looking to put cellular V2X (vehicle to everything) communication onto the next level, enabling vehicles to talk to other vehicles, to infrastructure, and to networks.
- **Healthcare:** The healthcare segment stands to benefit from 4G overall in a lot of ways when we bring IoT into play, there's a number of interesting use cases that come to the fore. Wearables are predicted to decrease hospital costs by 16% in the next five years. With 5G, remote monitoring, as well as faster data processing, can show patients and doctors healthcare data from wearables and insights based on it in near real time.

While it might be great to understand all the benefits we stand to gain out of bringing IoT onto 5G, there are certain requirements to be satisfied for that to happen as well.

WHAT IT WOULD NEED TO SUCCEED

The Internet of Things, to be successfully implemented and operated on 5G networks, puts forth its own set of requirements.

- **Standardisation:** For IoT devices to be able to talk to each other and

the internet, most of them currently need a central hub or some kind of intermediary. Technology like LPWAN will make these pointless to a large extent, opening the door for better inter-manufacturer standardisation, which is essential for the 5G-IoT combination to work well.

- **Wireless power:** So far, most IoT devices depend on batteries or wires for power. The former can prove to be an issue for devices in segments like security or manufacturing, which need to work 24x7 and can encounter durability or reliability issues due to battery failure. On the other hand, with more IoT devices incoming, wires are definitely going to be a problem as a power source due to the clutter they will create. The 5G-IoT combination needs wireless power, without a pad, for industrial as well as home IoT to work seamlessly.
- **Establish recycling:** if the total number of IoT devices scales up, so will the number of devices that fail. E-waste is a growing concern and future IoT growth will only add to that unless a proper recycling strategy is developed.

Even with all these three aspects in place, in a robust and reliable form, there's a fourth aspect that is perhaps the most important factor in the 5G-IoT combination – security.

THE SECURITY QUESTION

According to Nokia's Threat Intelligence Report 2019, IoT botnet activity represented 78 percent of malware detection events in communication service provider networks in 2018. In terms of growth rate, this is more than double the rate in 2016, which was when IoT botnets made their presence felt for the first time.

Problems with individual use cases are quite predictable. If 5G will boost IoT applications in automotive, healthcare and home security segments, it can also create risks of breaches of data and control in all of those segments. We need to look one level deeper to identify security issues that are more significant. Upcoming cloud and virtualisation technologies that are looking forward to 5G to be success-

fully deployed are also vulnerable to security issues on 5G. Software-defined networking (SDN) and network functions virtualization (NFV) are two such technologies whose USP is their open, flexible, programmable nature, and hence have to be implemented at the core of 5G implementations as in isolation they would be vulnerable to their own advantages.

Network slicing is another aspect of 5G that influences network security in a unique way. Since a 5G network can be spliced into slices that are purposed separately, each slice can have its own security implementation as per its requirements. For instance, a 5G network connection to a smart home could be connected to the smart door lock as well as an ambient light sensor. The former would be used occasionally whereas the latter would require constant connectivity, and hence could be put on two different splices with two different security implementations.

There might be issues with the 5G standard itself. In a paper titled "A Formal Analysis of 5G Authentication," researchers from ETH Zurich, the University of Lorraine and the University of Dundee, researchers found 5G Authentication and Key Agreement (a system for enabling networks to trust each other, used by 3G and 4G as well) has at least two major vulnerabilities. First, it enables one malicious user to move usage charges to another user. Second, it's possible to find nearby phones, which enables tracking of other users.

CONCLUSION

The numerous ways in which we stand to benefit from bringing 5G and IoT together will be rendered moot if it is done hastily. We've chosen to focus on the threats in the last section to highlight what could go wrong if the security features that are detailed in the 5G standard are not implemented properly. The increasing invasion of IoT devices that 5G will bring, along with their usage in mission-critical situations, makes security paramount. As most security researchers have put it, deploy 5G securely or don't deploy it at all. 